DevOps



LAB 04

Simple Python Database Connections

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BCS 7-B

Department Of Computer Science

Task 1: Fork and Clone the Databases Repository

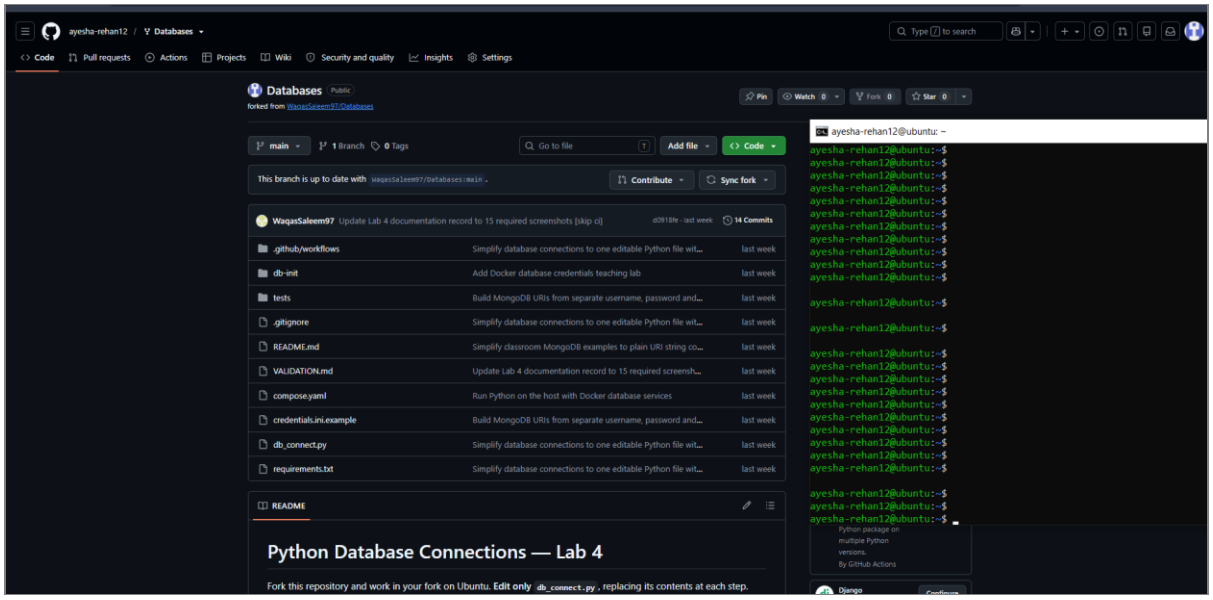


Figure: Fork of the Databases repository showing ayesha-rehan12/Databases and the original repository as its source.

```
ayesha-rehan12@ubuntu:~$ cd ~
ayesha-rehan12@ubuntu:~$ git clone git@github.com:ayesha-rehan12/Databases.git
Cloning into 'Databases'...
remote: Enumerating objects: 101, done.
remote: Counting objects: 100% (101/101), done.
remote: Compressing objects: 100% (90/90), done.
remote: Total 101 (delta 30), reused 1 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (101/101), 39.86 KiB | 27.00 KiB/s, done.
Resolving deltas: 100% (30/30), done.
ayesha-rehan12@ubuntu:~$ cd ~/Databases
ayesha-rehan12@ubuntu:~/Databases$ git remote -v
origin  git@github.com:ayesha-rehan12/Databases.git (fetch)
origin  git@github.com:ayesha-rehan12/Databases.git (push)
ayesha-rehan12@ubuntu:~/Databases$
```

Figure: Fork cloned over SSH on the Ubuntu server.

Task 2: Install Docker, Docker Compose, and Python on Ubuntu

```
Last login: Sun Oct 25 18:50:57 2026 from 192.168.1.25
ayesha-rehan12@ubuntu:~$ git --version
git version 2.53.0
ayesha-rehan12@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ayesha-rehan12@ubuntu:~$ docker compose version
Docker Compose version v5.6.1
ayesha-rehan12@ubuntu:~$
```

Figure: git --version, docker --version, and docker compose version showing installed versions.

Task 3: Start the Database Services

```
(.venv) ayesha-rehan12@ubuntu:~/Databases$ docker compose up -d --wait mysql mongo-dev mongo-test mongo-prod
[+] up 6/8
 Image mongo:7.0                 Pulled                                                                    382.5s
 Network databases_default       Created                                                                     0.7s
 Container databases-mysql-1     Healthy                                                                   166.7s
 Container databases-mongo-test-1 Healthy                                                                  192.2s
 Container databases-mongo-prod-1 Healthy                                                                  183.2s
 Container databases-mongo-dev-1  Healthy                                                                  182.2s
(.venv) ayesha-rehan12@ubuntu:~/Databases$ docker compose ps
NAME                    IMAGE       COMMAND                  SERVICE      CREATED         STATUS                   PORT
S
databases-mongo-dev-1    mongo:7.0   "docker-entrypoint.s…"   mongo-dev    3 minutes ago   Up 3 minutes (healthy)   127.
0.0.1:27017->27017/tcp
databases-mongo-prod-1   mongo:7.0   "docker-entrypoint.s…"   mongo-prod   3 minutes ago   Up 3 minutes (healthy)   127.
0.0.1:27019->27017/tcp
databases-mongo-test-1   mongo:7.0   "docker-entrypoint.s…"   mongo-test   3 minutes ago   Up 3 minutes (healthy)   127.
0.0.1:27018->27017/tcp
databases-mysql-1        mysql:8.4   "docker-entrypoint.s…"   mysql        3 minutes ago   Up 3 minutes (healthy)   3306
0/tcp, 127.0.0.1:3307->3306/tcp
(.venv) ayesha-rehan12@ubuntu:~/Databases$
```

Figure: All four database services (mysql, mongo-dev, mongo-test, mongo-prod) running and healthy.

```
0/tcp, 127.0.0.1:3307->3306/tcp
(.venv) ayesha-rehan12@ubuntu:~/Databases$ docker compose port mysql 3306
127.0.0.1:3307
(.venv) ayesha-rehan12@ubuntu:~/Databases$ docker compose port mongo-dev 27017
127.0.0.1:27017
(.venv) ayesha-rehan12@ubuntu:~/Databases$ docker compose port mongo-test 27017
127.0.0.1:27018
(.venv) ayesha-rehan12@ubuntu:~/Databases$ docker compose port mongo-prod 27017
127.0.0.1:27019
(.venv) ayesha-rehan12@ubuntu:~/Databases$
```

Figure: Host ports used by Python for each database service.

Task 4: MySQL with Values in the Code

```
127.0.0.1:27019
(.venv) ayesha-rehan12@ubuntu:~/Databases$ nano db_connect.py
(.venv) ayesha-rehan12@ubuntu:~/Databases$ python db_connect.py
Connected to MySQL: school_mysql
Server: 127.0.0.1 3307
Environment: dev
(.venv) ayesha-rehan12@ubuntu:~/Databases$
```

Figure: Connected to MySQL: school_mysql on 127.0.0.1 3307 with Environment: dev.

Task 5: MongoDB with Values in the Code

```
(.venv) ayesha-rehan12@ubuntu:~/Databases$ nano db_connect.py
(.venv) ayesha-rehan12@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) ayesha-rehan12@ubuntu:~/Databases$
```

Figure: Connected to MongoDB school_dev on port 27017 with environment dev.

Task 6: MongoDB with os.getenv()

```
ayesha-rehan12@ubuntu: ~/Databases
(.venv) ayesha-rehan12@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) ayesha-rehan12@ubuntu:~/Databases$
```

Figure: Environment-variable version connected to school_dev on port 27017 (dev).

```
ayesha-rehan12@ubuntu: ~/Databases
(.venv) ayesha-rehan12@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
(.venv) ayesha-rehan12@ubuntu:~/Databases$
```

Figure: Same code connected to school_test on port 27018 (test).

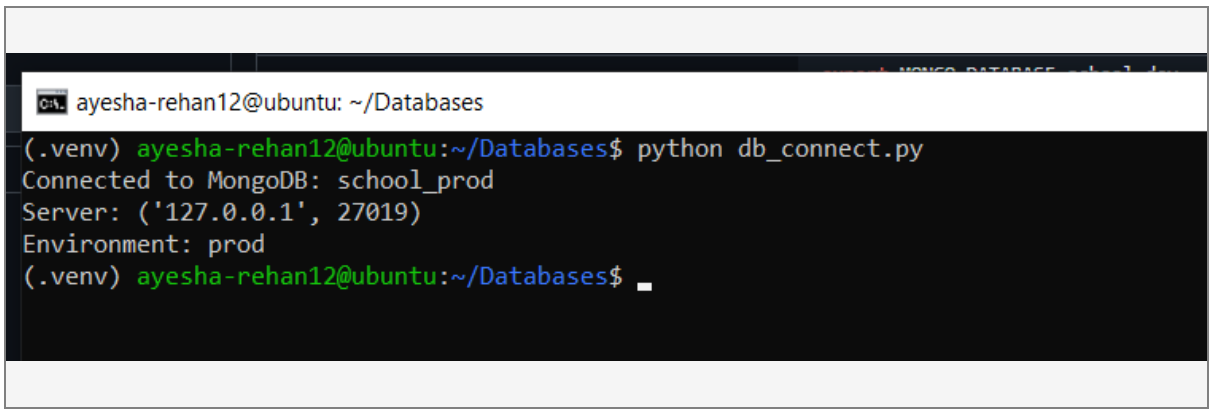


Figure: Same code connected to school_prod on port 27019 (prod).

Task 7: MongoDB with a Credentials File



Figure: credentials.ini created from credentials.ini.example.



Figure: credentials.ini restricted with chmod 600 (mode 600 shown).

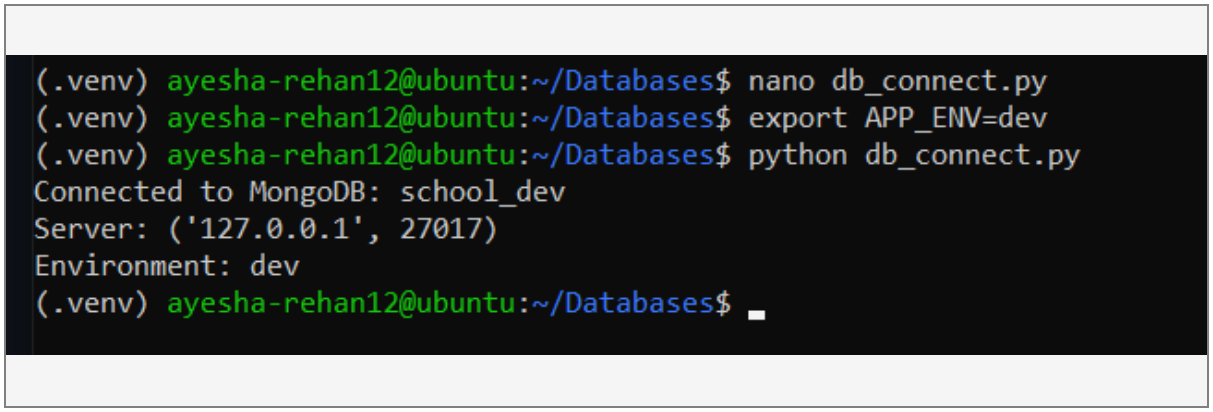


Figure: Credentials-file version connected to dev using APP_ENV=dev.

```
(.venv) ayesha-rehan12@ubuntu:~/Databases$ export APP_ENV=test
(.venv) ayesha-rehan12@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
(.venv) ayesha-rehan12@ubuntu:~/Databases$
```

Figure: Credentials-file version connected to test using APP_ENV=test.

```
(.venv) ayesha-rehan12@ubuntu:~/Databases$ export APP_ENV=prod
(.venv) ayesha-rehan12@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod
(.venv) ayesha-rehan12@ubuntu:~/Databases$
```

Figure: Credentials-file version connected to prod using APP_ENV=prod.